RESEARCH + CONTROLLED ASSET EXPERIMENT · 23 OCTOBER 2026

Production-quality AI art and animation for a faithful HoMM3 remaster

Evidence, diagnosed failure modes, three executed Pikeman approaches, reproducible workflows and a recommended production process. Research candidates remain separate from production assets.

1. Findings and recommendation

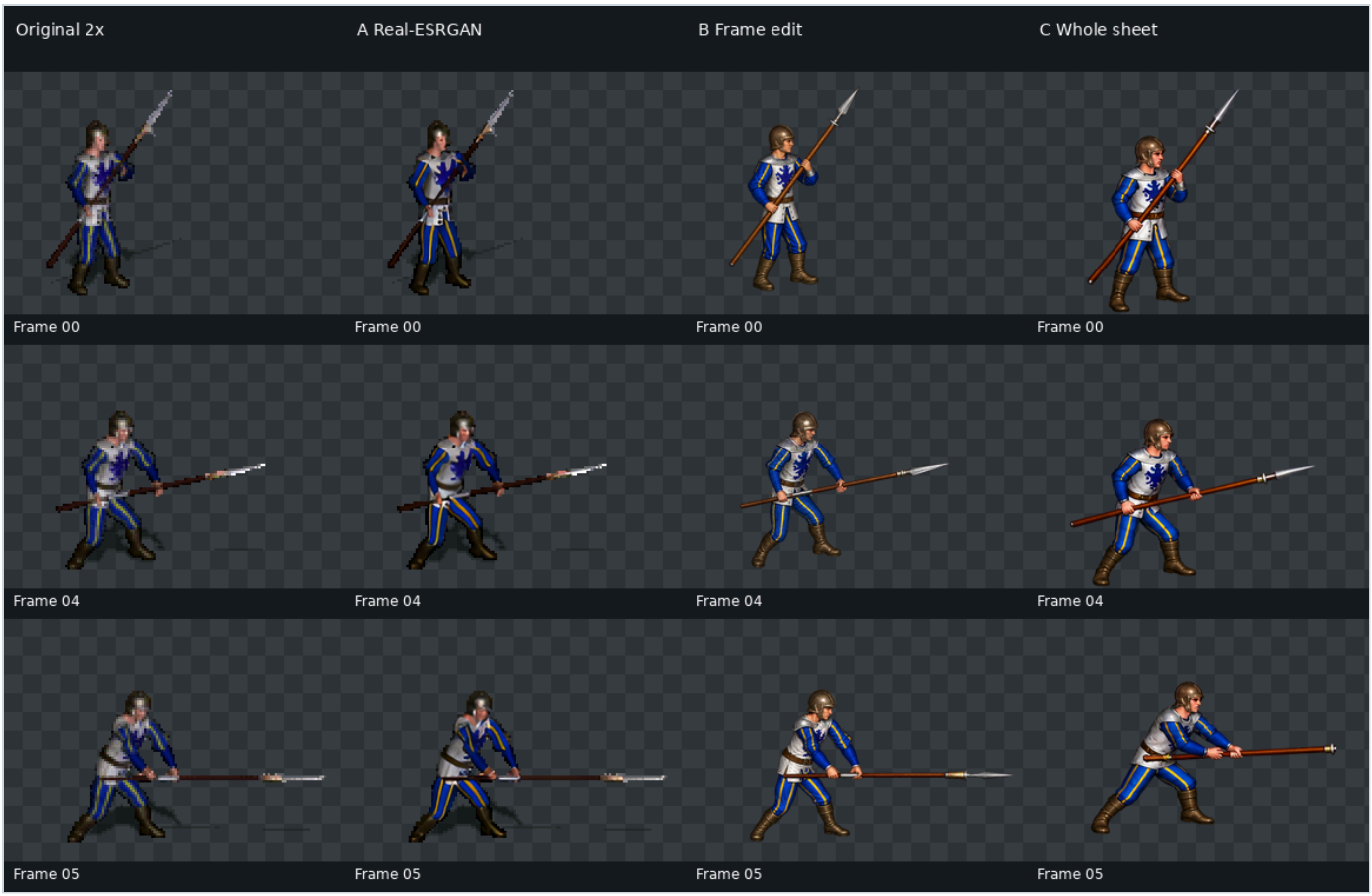
The strongest next pipeline is source-preserving restoration, artist-approved high-resolution keyframes, then structurally controlled editing or guided propagation with frame-level validation. Whole-sheet prompting should remain an experiment. None of the candidates tested here is approved for production.

The current workflow preserves many file-level properties, but it does not reliably preserve the original animation. A concrete example is Pikeman's recovery: the generated sheet contains ten frames, yet its last two poses correspond to the wrong source poses. Higher resolution alone cannot repair that error.

Three approaches were actually executed on one source group. Real-ESRGAN with locked source alpha preserved geometry and cleaned some surfaces. Independent reference edits added detail but altered silhouettes and character details. Whole-sheet editing produced ten recognizable figures, but changed the canvas, crossed cell boundaries and changed repeated poses. The named Qwen, ControlNet, temporal-video and rigging alternatives below were researched; they were not run.

> **Production decision:** retain the originals as coordinate and motion authority. Use the restoration baseline as a fidelity control. Validate a single creature with artist-approved structural guides before choosing a model or expanding the roster.

Approach	Executed result	Disposition
A · Restoration/upscale	10 frames, exact source alpha, exact repeat mapping; inferred surface detail	Useful baseline; human art review still required
B · Reference-guided frame editing	6 unique edits assembled into original 10-frame order; soft pose guidance; material and pose drift	Promising detail treatment; requires stronger structural control and cleanup
C · Whole-sheet editing	One 5×2 generated image; aspect and grid violations; pike crosses cells	Reject this raw output for atlas use
Controlled local Qwen / ControlNet / propagation / rig	Primary-source research and workflow handoff only	Next controlled trials; no measured project result



Same source, same fixed crop: ready (00), transition (04), full thrust (05). B/C are uncorrected drawings after the explicitly documented canvas packaging.

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2. Inspect the experimental animation

Frame indices are zero-based. All methods contain the same ten output positions and use the shared diagnostic timing: ten 100 ms dwells. Scrub to inspect pose correspondence. Cyan is the original opaque silhouette, magenta the candidate, white their overlap.

The PDF uses the still comparisons. The standalone HTML provides the synchronized scrubber.



Full-thrust opaque-mask comparison after fixed packaging; both masks threshold alpha > 200. Correspondence is not a measure of beauty.

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3. Project evidence and source limitations

Inspected `/workspace/Laema` at commit `16b4c3b7`. Git status was clean before and after the work. All new artifacts are under `/workspace/homm3-research`; the roster, production assets, project documents and gameplay code were not modified.

The checkout contains original DEF preservation files, indexed exports, palettes, decoded RGBA sheets and metadata. Its collection inventory records 2,912 DEF resources and 39,939 frames. This research did not rerun the entire collection's decoder verification. The selected CPKMAN group was inspected directly, and 32 current high-resolution Castle atlases were independently checked.

The external packages `castle-creature-assets-v01` and `pikeman-flipbook-assets-v02` are referenced in local provenance but are not present as complete master packages. Their source prompts, original high-resolution character masters and generation graphs could not be inspected. Connected Drive metadata searches for the package and Pikeman returned no results. The available runtime sheets were inspected; they must not be described as original master artwork.

B and C therefore share a clearly labeled **appearance proxy**: the first frame of the existing generated Pikeman idle atlas, tightly cropped to 141×222. This makes their reference input comparable, but also carries forward that generated design's biases. It is not an independent historical ground truth or proof of the missing master's quality.

Layer	Established limitation	Implication
Source art	The person in the selected ready frame is approximately 74 native pixels tall, excluding the raised weapon. Fine face, heraldry, fingers and surface detail are ambiguous.	A model can propose detail; it cannot demonstrate recovery of unavailable original high-resolution geometry.
Source encoding	DEF metadata supplies frame order, canvas and offsets. Preview alpha includes palette shadow interpretations (64/128) alongside opaque character pixels.	Keep indexed masters and shadow/selection semantics; do not flatten all channels into an RGB screenshot.
Timing	Pikeman provenance records a 1.00 s base attack duration from installed CrAnim.txt. DEFs do not provide per-frame dwell; battle speed and exact engine scheduling were not reconstructed.	Ten 100 ms dwells are a comparison assumption. Exact original timing remains unresolved.
Missing masters	Only selected generated runtime sheets and local provenance are available. Prior model settings and complete master packages are absent.	Diagnosis can establish output defects, not the exact historical cause or which model produced them.

Project sources: original collection README.md / README.txt; CPKMAN metadata.json; Pikeman PROVENANCE.md; repaint_v02/PROVENANCE.md. Detailed file and line references are in project_render_audit.md. The selected DEF SHA-256 is `f6fa622a5452b561beeb285bc234c14d728a301576357b9b41c6d0ae8b01b94c`.

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4. Why the current results depart from a faithful remaster

Source-art limitations

Large original canvases contain small rendered characters. Treating the 450×400 cell dimensions as the character's detail resolution overstates the available information. At large enlargement, source pixel steps and ambiguous

interior marks remain. Automatic human-pose detectors are also a poor default for tiny armor, pikes, wings and quadrupeds; use original-derived edges and manually verified landmarks instead. That detector warning is an engineering inference, not a detector benchmark performed here.

Generative limitations

More detailed figures can still change proportions, camera, limb placement and costume design. Existing Pikeman's full thrust is a deeper, wider lunge than its corresponding source pose. Archer's opening shooting frames simplify the original hand choreography. Griffin broadly preserves the action and looks substantially cleaner, but changes chest, crest, beak and claw proportions and omits the source's projected shadow. These examples do not establish gross anatomy failure across the roster, and they do not justify calling every generated frame unusable.

Workflow and consistency problems

Frame count preservation did not ensure pose correspondence. Original Pikeman group 12 has exact repeated pairs 1=9, 2=8, 3=7 and 4=6. Its current repaint instead repeats 0=9 and 1=8, while retaining 3=7 and 4=6. The recovery therefore raises the pike early. Full-cell RGBA hashes prove the repetition pattern; the contact sheet shows its visual effect. Conversely, Griffin's 0=8 and Archer's 3=4=5 correctly preserve source repeats.

A nominal atlas anchor does not make generated feet stationary. Likewise, coherent colors do not prove a rigid weapon, the same face, or the same pose. Keeping provenance, poses, duplicate identities, raw generations and cleanup history together is essential. No evidence in the available checkout establishes that the previous process used Qwen, ControlNet or a particular sampler.

Import, animation and rendering

The technical asset audit is reassuring: all 32 non-Pikeman highres_v01 atlases, containing 236 populated frame positions, match their recorded hashes and corresponding original group counts. No occupied alpha touches a cell edge; padding cells are empty. Imports use lossless mode, no size limit and no mipmaps. These checks exclude those specific corruption, count and clipping causes; they do not establish artistic fidelity.

Playback is deliberately retimed for Laema: Idle 8 fps, Walk 9 fps, Death 8 fps and Hurt distributed over a gameplay reaction duration. Several attacks use provisional one-second clocks. Griffin's nine-frame attack is sampled as 0,1,2,3,4,4,5,6,7,8, giving pose 4 an extra dwell. This is an objective runtime timing choice, not recovered original playback. Linear filtering and fractional display transforms can soften detail, but this research found no evidence of lossy texture import.

Several creatures retain provisional inherited collision or attack geometry; Archer's arrow and Monk's bolt remain original low-resolution sprites. These can weaken visual coherence independently of drawing quality. One tempting finding was rejected: Pikeman's provenance mentions a historical 115 px weapon/query gap, but current native geometry reaches the recorded tip extent. That old paragraph is stale and is not a current defect.



Existing Pikeman: original above generated within each comparison pair. One fixed body-height scale and root transform apply to the entire group; no per-frame alignment. Inspect recovery frames 08/09.



Existing Archer shooting group: compare opening hand positions and body proportions. Original 3/4/5 repeats are deliberately preserved.



Existing Griffin: broad motion correspondence is useful, while redesign and shadow differences remain. Cleaner detail alone does not establish 1:1 fidelity.

5. Practical techniques and strength of evidence

Primary repositories, model cards, reproducible examples and developer reports were checked on 23 October 2026. A publisher's static editing demo demonstrates that edit, not exact animation preservation. An issue report identifies a plausible failure mode, not its frequency.

Technique	Evidence and practical fit	Limits / execution here
ComfyUI	Official downloadable image-edit graphs provide a useful orchestration and provenance layer. Save graph, exact model hashes, node versions, inputs and outputs. Native 2511 example.	A graph is not a temporal consistency mechanism. Official template bundled; not executed.
Qwen-Image-Edit-2511	Apache-2.0 model card includes multi-reference runnable inference and character-consistency examples. Good next candidate for materials/details on a source-authoritative pose. Model card and code.	Static examples do not prove exact registration or a faithful sprite sequence. Not run.
Qwen-Image-2.1	September 2026 release with a 7B visual model, native RGBA and multiple references. Its Comfy edit templates wire ten reference slots. Official release, native workflow.	Research license restricts model use to noncommercial research/evaluation without a separate commercial license. Do not assume 2511's Apache terms carry over. License. Not run.
Qwen + ControlNet Union	InstantX publishes runnable canny, soft-edge, depth and pose examples; documented guidance 0.8–1.0, 30 steps, true CFG 4. Model and inference.	Documented base is Qwen/Qwen-Image, not 2511 or 2.1. Do not combine incompatible versions. No local ControlNet run.
FLUX.2 klein 4B	Permissive 4B variants offer multiple-reference editing and smaller local inference. Distilled four-step and Base workflows support different quality/speed tradeoffs. Official code, 4B card.	Verify the exact variant's license and measured GPU memory; broader family terms differ. No HoMM3 fidelity demonstration found. Not run.
SDXL + structural controls + IP-Adapter	Separately adjustable edge/lineart guidance and reference appearance are useful for a controlled comparison. Union model examples, Comfy implementation.	Older alternative, still practically tunable. Base and add-on licenses must be recorded. Not run.
Real-ESRGAN / RealESRNet	Open code and weights make restoration reproducible; RealESRNet is a useful less-adversarial alternative. Official implementation.	RealESRGAN_x4plus was actually run. RealESRNet and anime/video models were researched only; an anime label does not establish suitability for prerendered fantasy art.
Keyframe painting + EbSynth propagation	Open implementation includes example-based appearance transfer with guide pairs. Appropriate for propagating approved paint detail through a short original sequence. Implementation and examples.	Shape mismatch, occlusions and disocclusions require more keyframes or repainting. Official guidance. Not run.
TokenFlow / Rerender A Video	Feature correspondence and keyframe/patch propagation address time explicitly. Both publish code/examples. TokenFlow, Rerender A Video.	TokenFlow requires good inversion and warns of decoder jitter. Rerender documents a 24 GB GPU setup. Reconstruction must pass before edited results are evaluated. Not run.
Wan-Animate-2	August 2026 open release directly consumes a driving video, avoiding an intermediate pose extractor. Official code and examples.	A credible newer animation candidate; no evidence found for exact original sprite-frame, alpha or atlas compliance. Not run.

Technique	Evidence and practical fit	Limits / execution here
Sprite Sheet Diffusion	Paper and public code implement pose-guided temporal sprite generation. Paper, code.	Upstream open issue reports missing trained pose_guider.pth and incompatible legacy replacement. Do not call the released pipeline turnkey reproducible. Reproduction blocker. Not run.
ToonCrafter / interpolation	Official code demonstrates cartoon interpolation with optional sketch guidance. Code/examples.	Generates intermediate motion instead of retaining every original intermediate pose. Poor primary fit for strict 1:1 preservation; not run.

Implementation traps that matter here

Real-ESRGAN's helper ordinarily predicts alpha through the learned network, which can change thin silhouettes. In the inspected upstream CLI, the alpha_upsampler argument is parsed but not passed into enhance(); its alternate helper path labeled "bicubic" uses linear interpolation. Our executed baseline bypasses these ambiguities by running RGB restoration and explicitly reattaching source alpha. CLI source, helper source.

A developer report for Qwen-Image-Edit-2509 describes nonuniform pixel shifts/zoom in both Diffusers and ComfyUI; it is a relevant warning, not a benchmark of 2511 or 2.1. Current ComfyUI 2.1 documentation also discusses canvas-induced shifts, decoder checkerboards and downstream RGB/RGBA compatibility. Explicit output geometry still needs testing. Qwen issue, 2.1 documentation.

Temporal consistency cannot be established by a fixed seed or a good-looking sheet thumbnail. It requires correspondence, stable details, correct occlusion and playback review. Optical-flow warp error can help flag instability, but must exclude unreliable and occluded pixels; it is not anatomy or historical-fidelity evidence. RAFT implementation. No optical-flow metric was executed here.

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6. Controlled experiment: Castle Pikeman front attack

Selected group: CPKMAN group 12, block 11, ten ordered frames, six unique pixel drawings. Source full canvas 450×400. A single shared crop [150,172,296,276] gives 146×104 native pixels; guide images are nearest enlarged to 584×416. Candidate export canvases are 900×800 (2× source), with a 292×208 diagnostic crop. No motion interpolation or per-frame recentering was applied.

Timing qualification: all previews use ten 100 ms dwells, totaling the documented one-second base action. Exact original per-frame dwell and battle-speed behavior remain unverified. The experimental manifest preserves all source IDs and duplicate relationships so an authoritative timing schedule can later replace this assumption without regenerating images.

A. Restoration/upscaling baseline — executed

Ran official RealESRGAN_x4plus weights through the BasicSR RRDBNet forward path on CPU in FP32. Nearest valid RGB was bled under transparent pixels; the shared crop received ten reflected padding pixels. Native 4× RGB inference was downsampled to the 2× target. Original alpha was repeated exactly 2×; original semitransparent shadow RGB was retained. Nearest and Lanczos RGB controls used the same alpha policy. Six unique inputs were restored and repeated frames reused.

Execution took 18.20 seconds inside the script, excluding package/model download and installation. Every output retained source alpha exactly: zero changed alpha pixels across all 30 method/frame outputs. Restored surfaces look cleaner, while fine face/armor marks become simplified interpretations. The locked mask also retains source pixel steps; this is a deliberate fidelity tradeoff, not a complete high-resolution repaint.

Model SHA-256: `4fa0d38905f75ac06eb49a7951b426670021be3018265fd191d2125df9d682f1`. Real-ESRGAN commit `a4abfb2979a7bbff3f69f58f58ae324608821e27`; BasicSR commit `8d56e3a045f9fb3e1d8872f92ee4a4f07f886b0a`. Torch 2.14.1+cpu; Python 3.12.14. The compatibility shim removes registry imports and skips initialization before strict full weight loading; the network's forward source is unchanged. Full configuration and dependency freeze are included.

B. Reference-guided frame editing — executed with soft pose constraints

Submitted each of the six unique source poses with the same appearance proxy to the available image editing tool. Instructions explicitly prioritized source silhouette, camera, feet, limbs, hands and pike endpoints. Four source repeats were assembled from the exact corresponding edited pixels, producing the original ten-frame mapping.

This was source-image and instruction guidance, not a hard ControlNet or skeleton-constrained run. The image tool exposes no backend model version, seed or sampler. It must not be reported as a Qwen, FLUX or ComfyUI test. Prompts, inputs and raw outputs are retained; a future identical prompt need not reproduce identical pixels.

Requested canvas was 584×416. Five raw edits returned 1486×1058, one returned 1486×1059. They were resized as whole canvases to 292×208, without subject fitting or translation. Fine detail improved subjectively, but torso, arm and weapon silhouettes changed; heraldic motifs and metal grip/collar details vary. No conclusive extra-limb failure was found. The preserved repeats are a deterministic assembly success, not evidence that the generator independently preserved time.

C. Whole-sheet generation with the same reference — executed

Submitted the complete 5×2 original pose sheet and the same appearance proxy, requesting a 2920×832 image with ten 584×416 cells. The model returned 2117×743: approximately a 19% change in aspect ratio. Ten figures are recognizable and broad pose progression survives, but the fixed grid does not.

The full-thrust pike crosses from cell 05 into 06; cell 06's pike also crosses into 07. Raw alpha has 11 occupied pixels at cell 05's right edge, 10 at cell 06's left edge and 6 at its right edge. Equal-grid atlas extraction cuts weapons and imports fragments into neighbors. This is an objective production failure even if the raw sheet looks attractive.

For diagnostic playback, rounded equal grid cells were extracted, uniformly scaled to a shared 237×208 rectangle and placed at x=27 in a 292×208 slot. This one fixed transform preserves their aspect and exposes the layout drift. It does not recover valid source registration or repair missing weapon pixels. Raw files and packaging.json preserve this distinction. All four source duplicate pairs differ in C.



Raw C output with the requested equal grid overlaid. Cell-crossing weapons make ordinary fixed-grid extraction invalid; these red lines were added only for inspection.

Measurement	A Real-ESRGAN	B frame edits	C whole sheet
Output frame positions	10	10 (6 generated + 4 reused)	10 extracted; raw grid invalid
Mean opaque silhouette IoU	1.000 (mask imposed)	0.697	0.180 under fixed letterbox packaging
Opaque bottom displacement	0 px	−4 to +3 px	−8 to +14 px
Original repeated pairs preserved exactly	4 / 4	4 / 4 by caching	0 / 4
Requested raw image dimensions	Deterministic export	Not honored; aspect approximately retained	Not honored; aspect changed
Transparency	Exact source alpha and shadow	Valid transparent PNG; generated edges	Valid transparent PNG; cell crossings
Manual drawing cleanup performed	None	None	None

Metrics use alpha > 200 for both source and candidate, excluding source 64/128 shadow pixels. IoU measures shape/placement correspondence, not art quality; C's score is particularly sensitive to its failed layout and documented packaging. Bottom displacement is opaque support, not a manually labeled foot landmark. No high-resolution ground truth, perceptual quality score or optical-flow score was available. One creature and one generation per unique pose/sheet do not establish model-wide failure rates.

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7. Actual engine appearance and quality assessment

An isolated copy of the project rendered the real Gym scene, background, floor, player, camera and linear sprite filter. Physics settled the actors, then processing was frozen and each attack frame was explicitly sampled. The current repaint uses native scale 1; A/B/C use a single uniform scale 180/148 to match the original ready person's 74 native pixels to the existing 180-unit standing body. Original anchor (196,268) maps to crop anchor (92,192). No per-frame placement correction was applied. Alpha-edge RGB fixing does not change alpha.

These are **staged engine renders, not ordinary combat playtests**. They reveal the pixel baseline, edge treatment, background readability, pose and registration differences. They cannot establish attack feel, hit timing, interruptions, both facings or final visual acceptance. The engine is Godot 4.6.3 using Mesa software GL compatibility; the project targets 4.7/Forward Plus. An unchanged 4.7-only ObjectiveWidget property raises a UI error, leaving its box incomplete. All 55 requested captures saved successfully; shutdown logs also contain resource warnings. No production project import was run.



Actual Gym renderer, full-thrust frame 05. Detail crops share the same scene framing. C's missing pike tip follows the cell split failure, rather than texture import clipping.



Full Gym context, rendered in the isolated copy. The incomplete upper-right objective panel is the disclosed engine-version issue.

Criterion	Evidence / remaining judgment
Historical visual fidelity	A preserves source geometry by construction. B/C add interpretations; no original high-resolution ground truth was available. User must judge whether new materials belong to HoMM3.
Identity consistency	B's shared proxy and cached repeats help; motifs/prop fittings still vary between unique poses. C has a coherent general outfit but does not preserve repeated source drawings.
Anatomy	No gross extra-limb defect established in this selected test. Shoulder/torso proportions, hand placement and crouch changes need frame-level human review.
Motion accuracy	Existing repaint recovery correspondence is objectively wrong. B follows the correct cached source map but changes internal poses. C fails both duplicate identity and atlas extraction.
Transparency and shadow	A preserves palette-derived alpha/shadows. B/C have real transparency but omit/reinterpret source shadow content; inspect edges on dark, white and gameplay backgrounds.
Alignment	Fixed transforms reveal B bottom movement and substantial C layout drift. Automatic per-frame recentering would conceal the failure and must not be used as a quality metric.
Timing	All experimental previews verify 10×100 ms. This is a shared comparison contract; original executable dwell remains unknown.
Manual correction effort	No manual drawing cleanup was performed, so accepted-frame cleanup cost is unmeasured. Record artist minutes and rejection rate before estimating roster cost.
Gameplay appearance	Actual renderer comparisons exist. Complete combat, contact geometry, both directions and target 4.7 behavior still require validation.

8. Reproducible candidate workflows

Executed restoration

From this research directory, follow a-restoration/README.txt to install the pinned CPU runtime, official BasicSR/Real-ESRGAN commits and verified weights, then run:

```
runtime/bin/python scripts/run_a_restoration.py
python scripts/prepare_comparisons.py
```

The restoration is reproducible at the code/configuration level. Keep the model, architecture-source SHA, package versions and CPU settings together. Floating-point differences on another runtime may prevent byte-identical RGB, so verify alpha/count/order exactly and document numerical differences.

Next local frame-edit trial — prepared, not executed

1. Use the same six unique poses and source duplicate map. Replace the appearance proxy with a user-approved faithful high-resolution master and corrected keyframe silhouettes.
2. Start with the bundled untouched official ComfyUI Qwen-Image-Edit-2511 graph, pinned to workflow-template commit `67984637eefcd4825dcf5c134e39034785be1682`. Its defaults include a resolution-scaling node, an FP8 checkpoint and an optional four-step Lightning path; do not treat those defaults as an exact-pose workflow.
3. For the explicit Diffusers comparison, use true CFG 4, 40 steps and fixed seeds. Pad the 584×416 source to 608×416 with 12 pixels on each side before encoding, then crop that padding after saving the raw output. The documented pipeline rounds dimensions to latent multiples; a requested 584-pixel width can silently become 576. Explicit compatible canvases remove one avoidable shift source.
4. Test source-frame + master reference first, then add a separately verified structural workflow. The bundled Qwen-base + InstantX Union example is a different compatible architecture, not a ControlNet attachment to 2511. Preserve original-derived edges/weapon endpoints and record control strength.
5. Save raw output before any mask repair. A source-mask-constrained derivative may enforce the silhouette, but cannot repair misplaced eyes, hands or armor underneath it. Evaluate both.
6. Review at native gameplay size, enlarged view, overlay and playback. Correct failures individually; reuse exact repeated poses. Log artist minutes per accepted frame and full-group rejection rate.

The workflows folder includes the official graph, proposed parameters and scripts with execution status. Syntax preparation is not inference validation. This CPU environment had no GPU and initially no diffusion runtime, so named diffusion models were not loaded.

Next whole-strip trial — lower priority

Use exactly the same references and poses, fixed empty gutters, known output dimensions and frame IDs in an external manifest. Budget resolution per character, not per sheet. Reject occupied gutters, reordered poses, missing tips, incorrect duplicates or automatic per-cell scale changes before evaluating aesthetics. Repeat with multiple seeds only after one raw output meets the structural contract. C here has already failed that contract; hand-segmenting its weapons would add correction effort rather than validate automatic atlas generation.

Keyframe propagation trial — researched, not executed

Approve faithful ready, transition and full-thrust keyframes, propagate details with original frames as guide pairs, and add keyframes around limb crossings and newly visible surfaces. An illustrative open EbSynth invocation is:

```
ebsynth -style approved_keyframe.png   -guide original_keyframe.png original_targetframe.png   -
output propagated_targetframe.png
```

Preserve frame count, offsets and dwell externally. Compare propagation against A and the corrected keyframes; do not smooth or interpolate motion to hide texture instability. Use segmentation/position guides where the implementation supports them.

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9. When controlled rigging is a better investment

A persistent rig offers stronger control over character identity, rigid weapons, lighting, camera and render alignment. The shipped Dead Cells production account describes a 2D model sheet → 3D character/skeleton → authored poses → PNG sequence and normal-map workflow. This is credible production evidence for the approach, though its pixel-art aesthetic is not a demonstrated HoMM3 remaster result. Thomas Vasseur's production account.

For this project, calibrate a fixed camera and lighting to the source, build an approved character and rigid pike, then match a pose to every unique original drawing. Retain intentional source exaggerations instead of substituting generic mocap. Render exactly the original ordered poses and reuse repeats; preserve timing in the manifest. Use silhouette, head, hand, foot and weapon-tip overlays, followed by paintover where the render misses the historical look. This proposed adaptation has not been executed.

2D rigs can be economical for breathing and modest limb motion, but body turns, foreshortened thrusts, wing overlap and horse/rider occlusion need alternate artwork and manual draw-order changes. Spine's export controls make explicit frame/FPS/loop handling possible. Spine export documentation. A small open renderer illustrates the 3D-to-sprite technique, without claiming to be Motion Twin's tool. PixelArtPipeline.

Prefer a rig trial if per-frame corrections repeatedly change the same anatomy, armor or weapon, or if many actions share the same creature. Prefer painted controlled edits for a few short motions when faithful keyframes already exist. Choose using measured accepted-group artist time and fidelity, not raw generation throughput. Neither route removes modeling, animation or art-direction work.

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10. Recommended production pipeline and remaining human work

1. **Freeze the preservation contract.** Version original DEF/indexed data, palettes, frame IDs, group selection, offsets, duplicate map and authoritative playback schedule. Recover exact timing from original engine/data before calling the result a 1:1 remaster.

2. **Approve identity and style.** An artist defines faithful proportions, camera, helmet, face, emblem, cloth, steel, weapon and shadow rules from originals. High-resolution masters must be reviewed against those sources at gameplay scale.

3. **Keep the restoration control.** Export source-preserving nearest/Lanczos/Real-ESRGAN baselines with deliberate alpha treatment. They reveal whether later detail improves the visible result while moving away from source geometry.

4. **Author structural keyframes.** Trace or rig all unique source poses, including hand and pike endpoints. Keep a fixed group transform, source pivot and layer semantics. Use a shared rigid weapon model/layer when appropriate.

5. **Add detail selectively.** Trial permissive Qwen 2511 or another documented compatible edit/control stack; use approved keyframes and propagation to reduce repeated interpretation. Record full model/workflow provenance. Keep raw and cleaned variants separate.

6. **Apply deterministic assembly and checks.** Reuse duplicates, retain frame count/order/dwell, validate alpha, gutter emptiness, canvas and anchors, then pack the atlas. Do not let a model invent the export format.

7. **Review the entire motion in context.** Artist checks anatomy, identity, materials, occlusion, shadow, planted feet and weapon continuity. Technical review checks actual import, filter, pixel scale, both facings, hit/contact alignment, transitions and timing in the target engine.

8. **Measure one accepted creature before scaling.** Record rejected generations, cleanup minutes per unique frame, total accepted-group cost, source fidelity and user preference. Only then select a production method for more creatures.

Human art direction remains necessary for ambiguous historical detail, proportion/style decisions, consistent heraldry, anatomically convincing hands and faces, occlusion repair, weapon construction, shadows and the final judgment of whether it looks like HoMM3. Human animation review must reject changed poses and spacing even when individual frames look better.

Unresolved: complete high-resolution masters and previous generation workflows; exact original per-frame timing; local Qwen/ControlNet, propagation and rig experiments; target Godot 4.7 combat playback; both-facing edge/contact checks; accepted-frame cleanup cost; and your visual acceptance. No production asset replacement is recommended from this test.

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11. Evidence and reproduction files

The portable review ZIP contains this report, inputs/proxy, raw B/C generations, candidate PNGs, source/experiment manifests, comparison sheets, animated previews, audit evidence, proposed workflows and capture scripts. It excludes downloaded model weights, the Python environment and the copied game project/cache. Reproduction requires the project and the separately downloaded pinned model weights.

File / folder	Purpose
inputs/source_contract.json	Frame identity, source rectangles, crop and provisional timing
a-restoration/README.txt; config.json; metrics.json	Executed restoration recipe, hashes, execution and per-frame evidence
generation_manifest.json; b-edit/raw; c-strip/raw	Exact prompts, input proxy and untouched model results; backend version unavailable

File / folder	Purpose
packaging.json; experiment_metrics.json; preview_checks.json	Explicit normalization, pixel measurements and animation-duration readback
visual_audit/	Existing Pikeman/Archer/Griffin comparisons and duplicate-frame audit
project_render_audit.md; runtime-sheet-audit.json	Code references and 32-atlas structural audit
gameplay-preview/normalized/; README.md	Final fair-size engine captures, sequences, limitations and reproduction
workflows/	Official upstream graph and clearly unexecuted local-edit handoff
scripts/prepare_comparisons.py; scripts/build_report.py	Rebuild measurements, comparisons and standalone report

Research claims are linked beside the relevant discussion. Project observations use the local evidence files above. PNGs are the candidate asset format; GIF/WebP/MP4 are review derivatives. No final art, gameplay, submission or commercial-model-license clearance is granted by this report.

Prepared from the local project and cited primary sources. Tested outputs and researched-only techniques are labeled separately. No production roster changes.